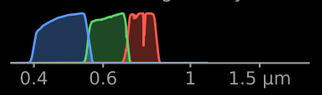
DISCOVER · wide-field surveys find ~10³–10⁴ lensed AGN

CHARACTERIZE

Rubin LSST

pixel 0.2″ · PSF 0.71–0.77″

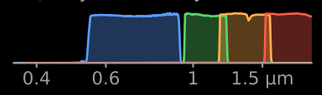
RGB = i r g · 10-yr



Euclid

pixel 0.1–0.3″ · PSF 0.16–0.5″

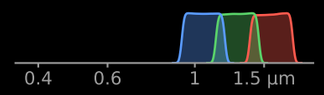
Q1 style: VIS + Y J H · wide



Roman

pixel 0.11″ · PSF 0.087–0.13″

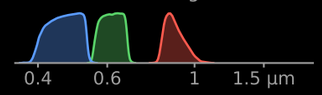
RGB = F158 F129 F106 · HLWAS



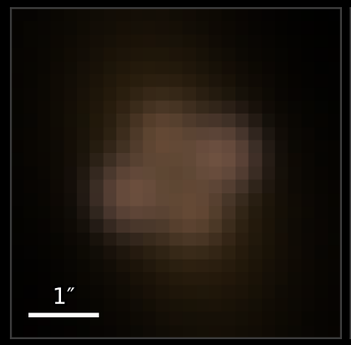
Lazuli WCC

pixel 17 mas · PSF 33–62 mas

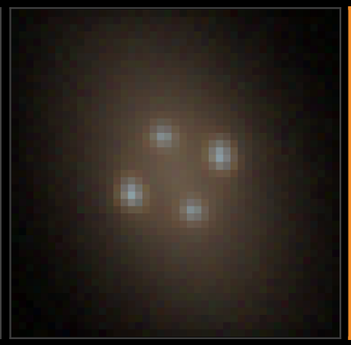
RGB = LZz LZr LZg · 4×300 s

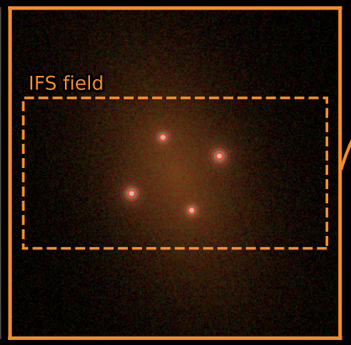


Quad z_d=0.43 z_s=1.87

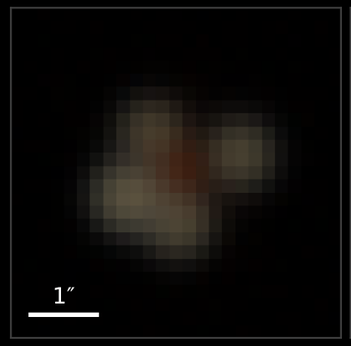




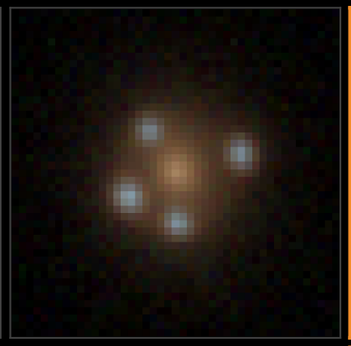


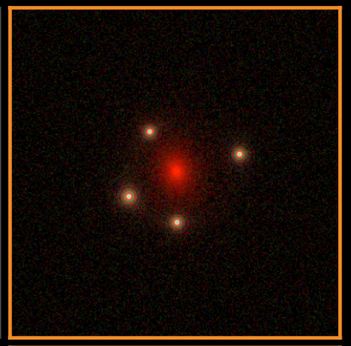


Quad z_d=0.89 z_s=2.74

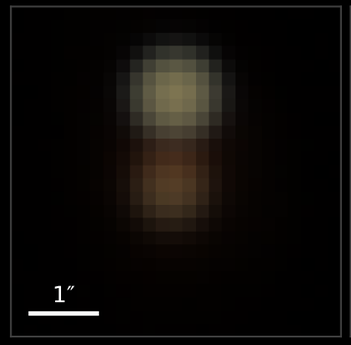


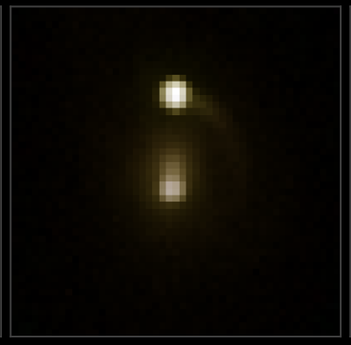


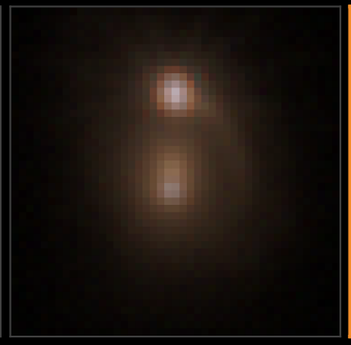




Double z_d=0.57 z_s=1.22



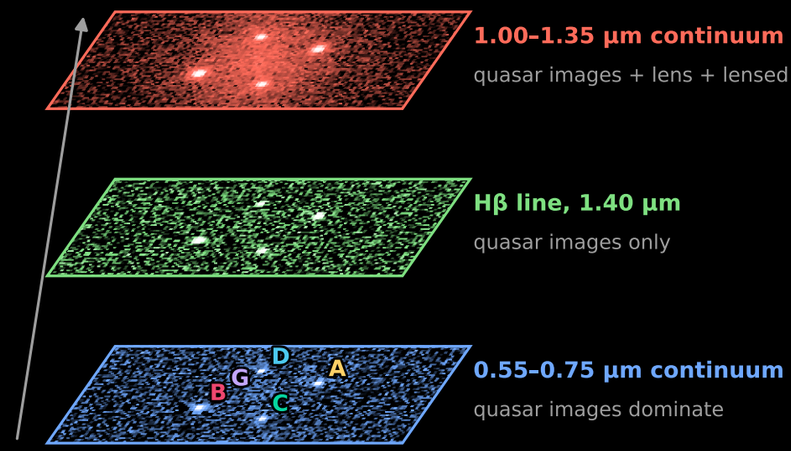






LAZULI IFS · one pointing, every image

0.4–1.7 µm · R ≈ 100–500 · 40 mas spaxels · 4 × 900 s

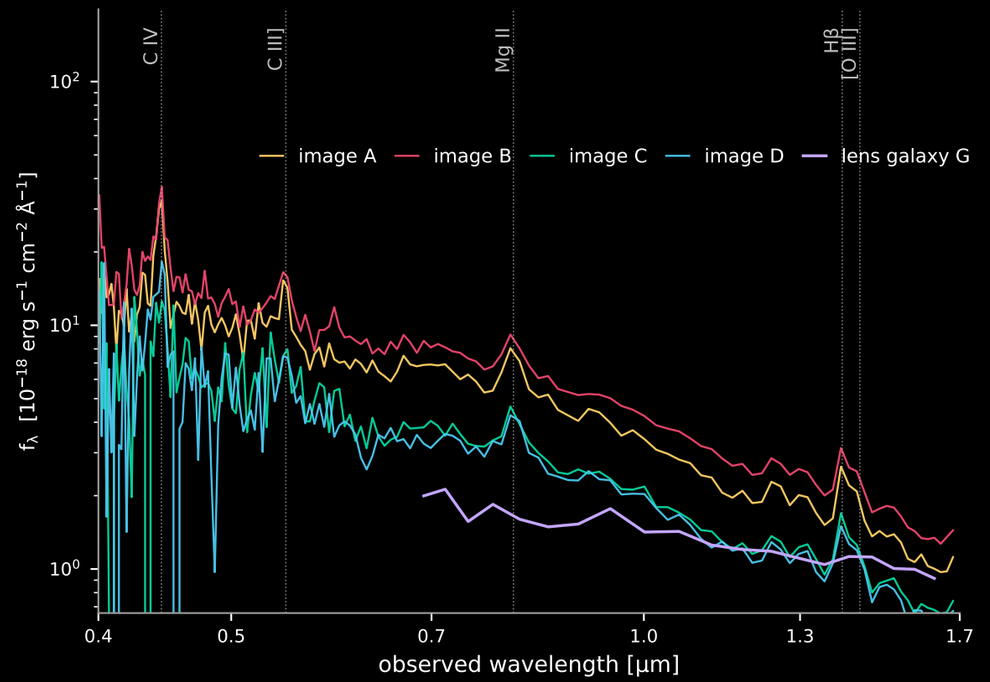


1.00–1.35 µm continuum
quasar images + lens + lensed host

Hβ line, 1.40 µm
quasar images only

0.55–0.75 µm continuum
quasar images dominate

Spectra of every image and the lens: source redshift and lensing confirmed in one visit



Simulated with SLSim: COSMOS hosts, qsogen quasar SEDs, static microlensing. Lazuli: 3.06 m, Strehl > 0.8 at 633 nm; specs from Roy et al. 2026 (arXiv:2601.02556). Lazuli throughput, sky, noise and exposure times are notional. Lensed-AGN counts: Oguri & Marshall 2010.